

LNP™ ELCRIN™ W1000XiQ

Polybutylene Terephthalate

SABIC

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Technical Data

Product Description

LNP ELCRIN W1000XiQ compound is based on Polybutylene terephthalate (PBT) resin utilizing ELCRIN iQ upcycling technology containing minimum 56% Post-Consumer Recycling (PCR) weight content. Added features of this grade include: Improved Processing.

General

Material Status	• Commercial: Active		
Literature ¹	• LNP ELCRIN iQ resin for circular solutions • LNP solutions for circular economy teaser • LNP™ ELCRIN™ iQ COMPOUNDS – SUPPORTING FOOD CONTACT COMPLIANCE		
Search for UL Yellow Card	• SABIC • LNP™ ELCRIN™		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Recycled Content	• Yes		
Features	• Chemical Resistant	• Good Processability	• Recyclable Material
Uses	• Appliances • Automotive Interior Parts • Decorative Parts	• Electrical Parts • Electronic Displays • Personal Care	• Sporting Goods
Processing Method	• Injection Molding		

Physical

	Nominal Value Unit	Test Method
Density / Specific Gravity	1.31 g/cm ³	ASTM D792 ISO 1183
Melt Mass-Flow Rate (MFR) (250°C/2.16 kg)	23 g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (250°C/2.16 kg)	21 cm ³ /10min	ISO 1133
Molding Shrinkage - Flow (3.20 mm)	1.6 to 2.4 %	Internal Method
Water Absorption		ISO 62
Saturation, 23°C	0.16 %	
Equilibrium, 23°C, 50% RH	0.070 %	

Mechanical

	Nominal Value Unit	Test Method
Tensile Modulus		
-- ³	2550 MPa	ASTM D638
--	2550 MPa	ISO 527-1/1
Tensile Strength		
Yield ⁴	56.0 MPa	ASTM D638
Yield	55.0 MPa	ISO 527-2/50
Break ⁴	29.0 MPa	ASTM D638
Break	50.0 MPa	ISO 527-2/50
Tensile Elongation		
Yield ⁴	3.2 %	ASTM D638
Yield	3.0 %	ISO 527-2/50
Break ⁴	140 %	ASTM D638
Break	130 %	ISO 527-2/50
Flexural Modulus		
50.0 mm Span ⁵	2560 MPa	ASTM D790
-- ⁶	2430 MPa	ISO 178
Flexural Stress		
-- ^{6, 7}	80.0 MPa	ISO 178
Yield, 50.0 mm Span ⁵	84.0 MPa	ASTM D790



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Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength ⁸ (23°C)	5.0 kJ/m ²	ISO 179/1eA
Notched Izod Impact		
-30°C	46 J/m	ASTM D256
23°C	50 J/m	ASTM D256
-30°C ⁹	2.0 kJ/m ²	ISO 180/1A
23°C ⁹	4.0 kJ/m ²	ISO 180/1A
Instrumented Dart Impact		ASTM D3763
23°C, Total Energy	61.0 J	

Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		
0.45 MPa, Unannealed, 4.00 mm, 64.0 mm Span ⁹	133 °C	ISO 75-2/Bf
1.8 MPa, Unannealed, 3.20 mm	48.0 °C	ASTM D648
Vicat Softening Temperature	168 °C	ISO 306/B120
CLTE		ASTM E831
Flow : -40 to 40°C	7.8E-5 cm/cm/°C	ISO 11359-2
Transverse : -40 to 40°C	8.1E-5 cm/cm/°C	

Injection	Nominal Value Unit
Drying Temperature	120 °C
Drying Time	3.0 to 4.0 hr
Suggested Max Moisture	0.020 %
Suggested Shot Size	40 to 80 %
Rear Temperature	230 to 250 °C
Middle Temperature	240 to 255 °C
Front Temperature	245 to 260 °C
Nozzle Temperature	240 to 255 °C
Processing (Melt) Temp	245 to 260 °C
Mold Temperature	50 to 75 °C
Back Pressure	0.300 to 0.700 MPa
Screw Speed	50 to 100 rpm
Vent Depth	0.013 to 0.025 mm

Injection Notes

- Drying Time (Cumulative): 12 hr

Notes

¹ These links provide you with access to supplier literature. We work hard to keep them up to date; however you may find the most current literature from the supplier.

² Typical properties: these are not to be construed as specifications.

³ 50 mm/min

⁴ Type I, 50 mm/min

⁵ 1.3 mm/min

⁶ 2.0 mm/min

⁷ at Yield

⁸ 80*10*4 sp=62mm

⁹ 80*10*4 mm



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Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

Amco Polymers

Telephone: 800-262-6685

Web: <http://www.amcopolymers.com/>

Availability: North America

Chase Plastic Services, Inc.

Chase Plastics Services is a North American distributor with representatives throughout the region. Please find your rep here: <http://www.chaseplastics.com/contact/locations>

Telephone: 800-232-4273

Web: <http://www.chaseplastics.com/>

Availability: North America

Guangzhou Hua Xiu Plastics Co. Ltd

Telephone: +86-20-82582555

Web: <http://www.va-so.cn/>

Availability: Asia Pacific

Nexeo Plastics

Nexeo Plastics is leading global resin distributor with the technical resources you need to overcome material challenges. Visit us on the web at www.nexeoplastics.com.

Telephone: 833-446-3936

Web: <https://www.nexeoplastics.com/>

Availability: North America

